

PRELIMINARY

**GOSFORD HIGH SCHOOL
MATHEMATICS**

Assessment task 3: June 2010

Paper 1

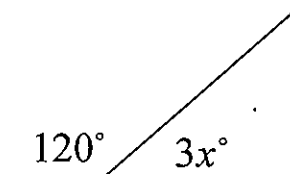
Start each question on a new page

Time: 60 minutes.

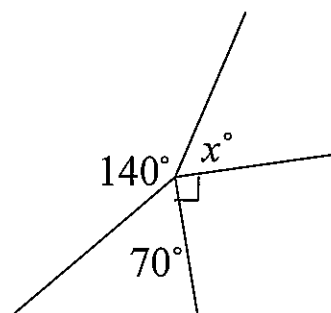
Question 1. (2 marks each)

Find the value of the pronumeral. (no reasoning is required).

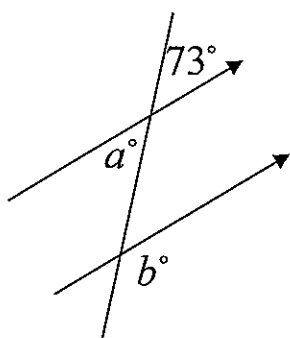
a)



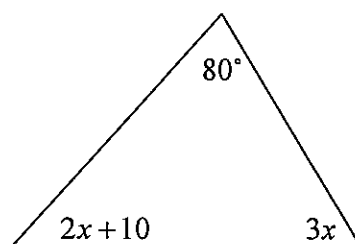
b)

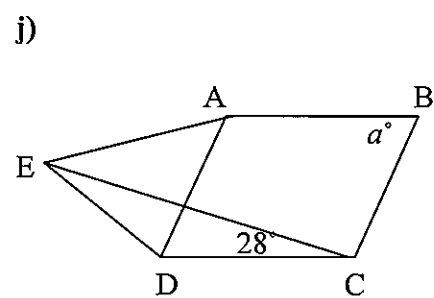
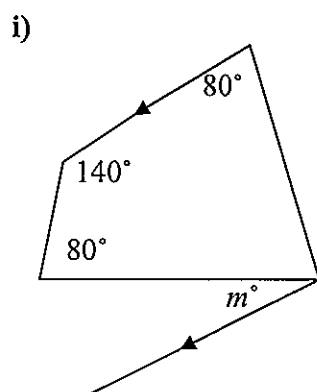
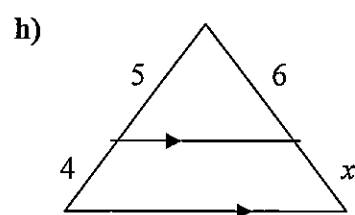
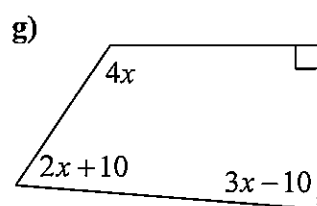
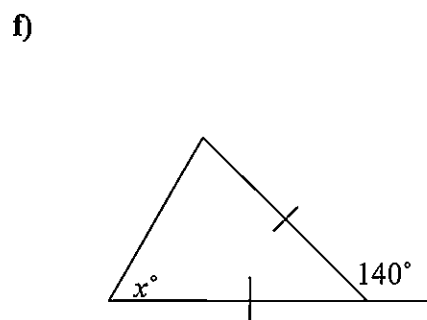
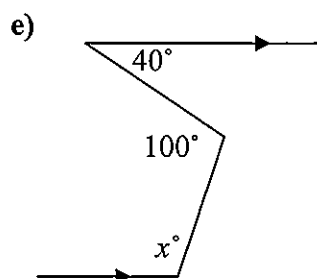


c)



d)

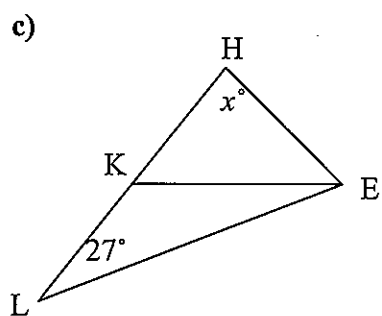
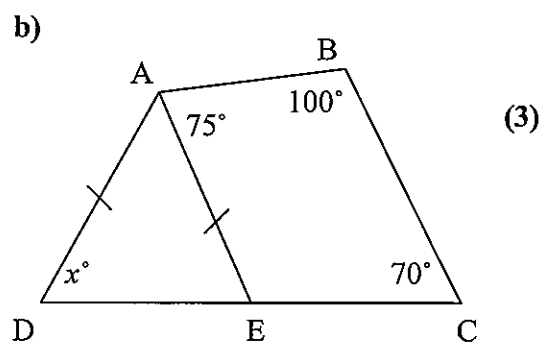
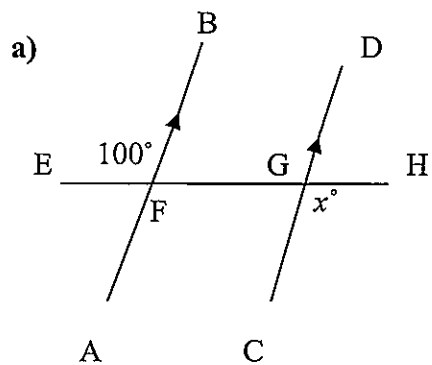




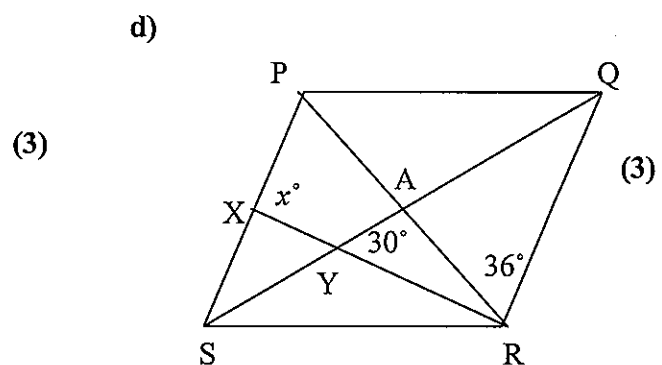
ABCD is a rhombus
 $\triangle DEA$ is equilateral

Question 2. (new page)

Find the value of the pronumeral giving full reasoning for your answer.

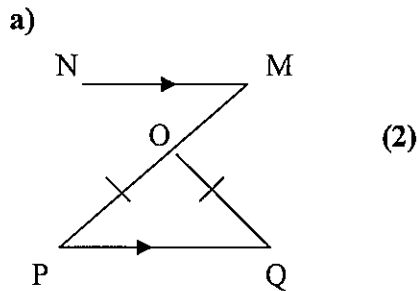


Given $HK = HE$
and $KL = KE$

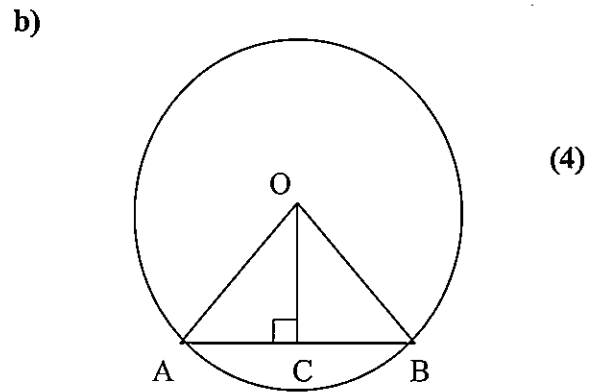


Given PQRS is a rhombus.

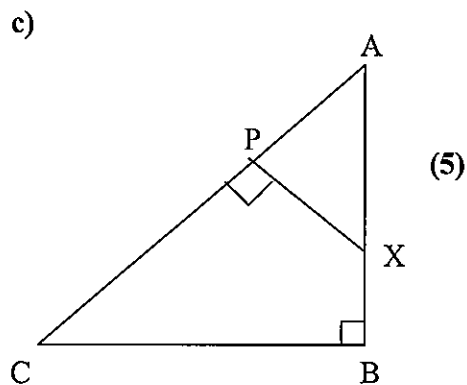
Question 3 (new page)



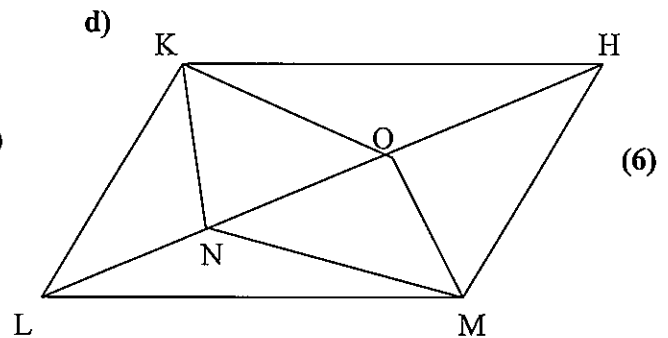
Given NM is parallel to PQ
and $OP = OQ$. Prove
angle NMO equals angle OQP.



If O is the centre of the circle and OC
is perpendicular to AB. Prove, by use
of congruent triangles, that OC
bisects AB.



- i) Show that triangle PAX is similar to triangle BAC.
- ii) If AX and BC are both 5cm and AP is 4cm find PX and AC.



Given HKLM is a parallelogram and
 $OH = HM$ and $LN = LK$.

- i) Show triangle LKN is congruent to triangle OHM.
- ii) Hence show KNMO is a parallelogram.

Question 4. (new page)

- a) If α and β are the roots of the quadratic equation $x^2 - 3x - 5 = 0$ find the value of:

i) $\alpha + \beta$
ii) $\alpha \beta$
iii) $\alpha^2 \beta + \alpha \beta^2$
iv) $\alpha^2 + \beta^2$
v) $\left(\alpha - \frac{1}{\beta}\right)\left(\beta - \frac{1}{\alpha}\right)$

(8)

- b) Write a quadratic equation whose roots are:

$3 + \sqrt{2}$ and $3 - \sqrt{2}$

(2)

- c) Solve

i) $x^4 - 3x^2 - 4 = 0$
ii) $9^x - 12 \cdot 3^x + 27 = 0$

(6)

- d) For the quadratic equation $x^2 - (k - 4)x + 3k = 0$ find the value of k if the product of the roots is twice the sum of the roots.
- (2)

- e) Find the value of A , B and C so that:

$$2x^2 - 7x - 4 \equiv A(x + 2)^2 + B(x + 2) + C.$$

(3)

- f) By using the substitution $m = \frac{x^2}{x+1}$, or otherwise, solve

$$\frac{x^2}{x+1} = 2 - \frac{x+1}{x^2}$$

(4)

GOSFORD HIGH SCHOOL
MATHEMATICS
Assessment task 3: June 2010
Paper 2

Start each question on a new page

Time: 60 minutes.

Question 1.

For the points $A(0,2)$, $B(5,4)$ and $C(1,-1)$

- i) Plot their coordinates on a number plane. (2)
- ii) Find the exact length of the interval joining B to C . (2)
- iii) Find the midpoint of the interval joining A to C . (2)
- iv) Find the gradient of the line passing through the points B and C . (2)
- v) Show that the equation of the line passing through the points B and C is $5x - 4y - 9 = 0$. (2)
- vi) Find the x intercept of the line in part (v) (1)
- vii) Find the perpendicular distance from A to the line $5x - 4y - 9 = 0$. (2)
- viii) Find the coordinates of D , such that $ABCD$ forms a parallelogram. (1)
- ix) Find the area of the parallelogram $ABCD$. (1)
- x) If A is the mid point of the points C and E , find the coordinates of E . (2)
- xi) Find the equation of the line that passes through A and makes an angle of 135° with the positive direction of the x axis. (2)

Question 2.(start a new page)

- a) Find the equation of the line that passes through the point (2,3) and is parallel to the line $y = 3x - 4$. (2)
- b) Find the equation of the line, in general form, that passes through the point (2,-1) and is perpendicular to the line $5x - 3y + 9 = 0$. (4)
- c) Find the equation of the line passing through the intersection of the lines whose equations are $3x - 4y - 1 = 0$ and $2x + 3y - 5 = 0$ and the point (1,2). (3)

Question 3. (start a new page)

- a) Show that the line $3x + 4y - 20 = 0$ is a tangent to the circle $x^2 + y^2 = 16$. (3)
- b) Find the shortest distance between the lines $3x - 4y + 9 = 0$ and $3x - 4y - 20 = 0$. (3)

Question 4. (start a new page)

- a) Find the value of a and b if $3\sqrt{5} - \frac{1}{\sqrt{5}-2} = a + b\sqrt{5}$ (3)
- b) Solve $x^2 \leq 5x$ (3)
- c) Solve $|x-1| \geq 4$ (2)
- d) Factorise $a^2 - b^2 + 2bc - c^2$ (2)
- e) Find x and y if $\frac{4^x}{16} = 8^{x+y}$ and $2^{2x+y} = 128$ (4)

ASSESSMENT TASK 3: PAPER 1

Q1) a) $3x = 60$
 $x = 20^\circ$

b) $x = 360 - (90 + 70 + 140)$
 $= 60^\circ$

c) $a = 73^\circ, b = 107^\circ$

d) $2x + 10 + 3x = 100$
 $5x = 90$
 $x = 18$

e) $x + 60 = 180$
 $x = 120^\circ$

f) $x + x = 140$
 $x = 70^\circ$

g) $9x = 270$
 $x = 30^\circ$

h) $\frac{x}{6} = \frac{4}{5}$
 $x = 4 \frac{4}{5}$

i) $60 + m + 80 = 180$
 $m = 40^\circ$

j) $28 + 28 + 60 + a = 180$
 $a = 64$

Q2) a) $\angle DGF = 100^\circ$
 (corresponding angles in parallel lines $AB \parallel CD$)
 $x = 100^\circ$ vertically opposite angles are equal.

b) $\angle AEC = 115^\circ$
 angle sum of a quadrilateral $= 360^\circ$.
 $\angle AED = 65^\circ$ straight angle.

$x = 65^\circ$ base angles of an isosceles Δ are equal

c) $\angle LEK = 27^\circ$
 base angles of an isosceles Δ are equal
 $\angle HKE = 54^\circ$
 exterior angle of a triangle equals the

sum of the two opposite interior angles.

$\angle HEK = 54^\circ$ base angles of an isosceles triangle are equal.
 $x = 72^\circ$ angle sum of a $\Delta = 180^\circ$

d) $\angle XYA = 150^\circ$
 straight angle.
 $\angle XPA = 36^\circ$

alternate angles in parallel lines $PS \parallel OR$ opposite sides of a rhombus $\angle PAY = 90^\circ$ diagonals of a rhombus intersect at right angles.

$x = 84^\circ$ angle sum of a quadrilateral $= 360^\circ$

Q3)

a) $\angle NMO = \angle OPQ$
 alternate angles in parallel lines.
 $\angle OPQ = \angle OQP$
 base angles of an isosceles Δ are equal
 $\therefore \angle NMO = \angle OQP$.

b) $\Delta OCA, \Delta OCB$
 $OA = OB$ radii of same circle
 OC common

$\angle OCA = \angle OCB = 90^\circ$
 $\therefore \Delta OCA \cong \Delta OCB$ R.
 $\therefore AC = BC$

Corresponding Sides in congruent Δ 's
 $\therefore C$ mid pt of AB
 $\therefore OC$ bisects AB .

c) i) $\triangle PAX, \triangle BAC$

$\angle A$ Common

$\angle APX = \angle ABC$

given

$\therefore \triangle PAX \parallel \triangle BAC$ (AAA)

$$\text{ii) } PX^2 = 5^2 - 4^2$$

$$PX = 3$$

$$\frac{AC}{5} = \frac{5}{3}$$

$$AC = \frac{25}{3}$$

d) $KL = HM$

opposite sides of a parallelogram are equal.

$\therefore LN = OH$

$\angle KLN = \angle MHO$

alternate angles in parallel lines $KL \parallel HM$.

$\therefore \triangle LKN \cong \triangle OHM$ (SAS)

ii) $KN = OM$

corresponding sides in congruent $\triangle$'s.

$\angle LNK = \angle HOM$ (congruent $\triangle$'s)

$\therefore \angle KNO = \angle MON$

straight angle

$\therefore KN \parallel OM$ alternate angles equal.

$\therefore KNMO$ parallelogram
opposite sides equal and parallel.

Q4)

$$a) x^2 - 3x - 5 = 0$$

$$i) \alpha + \beta = 3$$

$$ii) \alpha\beta = -5$$

$$iii) \alpha^2\beta + \alpha\beta^2 = \alpha\beta(\alpha + \beta)$$

$$= -5 \times 3$$

$$= -15$$

$$iv) \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 9 + 10 = 19$$

$$v) \left(\alpha - \frac{1}{\beta}\right)\left(\beta - \frac{1}{\alpha}\right)$$

$$= \alpha\beta - 1 - 1 + \frac{1}{\alpha\beta}$$

$$= -5 - 2 - \frac{1}{5}$$

$$= -7\frac{1}{5}$$

$$b) x^2 - (3 + \sqrt{2} + 3 - \sqrt{2})x + (3 + \sqrt{2})(3 - \sqrt{2}) = 0$$

$$= x^2 - 6x + 7 = 0$$

$$c) i) x^4 - 3x^2 - 4 = 0$$

$$(x^2 - 4)(x^2 + 1) = 0$$

$$x^2 = 4, x^2 = -1$$

$$x = \pm 2 \quad \text{no solution}$$

$$ii) 9^x - 12 \cdot 3^x + 27 = 0$$

$$(3^2)^x - 12 \cdot 3^x + 27 = 0$$

$$(3^x)^2 - 12 \cdot 3^x + 27 = 0$$

$$\text{let } a = 3^x$$

$$a^2 - 12a + 27 = 0$$

$$(a - 9)(a - 3) = 0$$

$$a = 9, a = 3$$

$$3^x = 9, 3^x = 3$$

$$\therefore x = 2, 1$$

$$d) x^2 - (k - 4)x + 3k = 0$$

$$2\beta = 2(\alpha + \beta)$$

$$3k = 2(k - 4)$$

$$3k = 2k - 8$$

$$k = -8$$

$$e) 2x^2 - 7x - 4 = A(x+2)^2 + B(x+2) + C$$

equating co-efficient of x^2 : $A = 2$.

$$\text{let } x = -2: 18 = C$$

$$\text{let } x = -1: 5 = A + B + C$$

$$5 = 2 + B + 18$$

$$B = -15$$

$$\therefore A = 2, B = -15, C = 18$$

$$f) m = 2 - \frac{1}{m}$$

$$m^2 = 2m - 1$$

$$m^2 - 2m + 1 = 0$$

$$(m - 1)^2 = 0$$

$$m = 1$$

$$\frac{x^2}{x+1} = 1$$

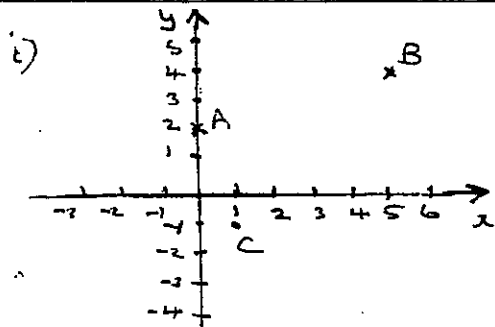
$$x^2 = x + 1$$

$$x^2 - x - 1 = 0$$

$$x = \frac{1 \pm \sqrt{1+4}}{2}$$

$$x = \frac{1 \pm \sqrt{5}}{2}$$

Assessment task 3. PAPER 2.



$$\begin{aligned} \text{ii) } BC &= \sqrt{(5-1)^2 + (4+1)^2} \\ &= \sqrt{16+25} \\ &= \sqrt{41} \end{aligned}$$

$$\begin{aligned} \text{iii) } &\left(\frac{0+1}{2}, \frac{2-1}{2}\right) \\ &= \left(\frac{1}{2}, \frac{1}{2}\right) \end{aligned}$$

$$\begin{aligned} \text{iv) } m &= \frac{4+1}{5-1} \\ &= \frac{5}{4} \end{aligned}$$

$$\begin{aligned} \text{v) } y-4 &= \frac{5}{4}(x-5) \\ 4y-16 &= 5x-25 \\ 5x-4y-9 &= 0 \end{aligned}$$

$$\begin{aligned} \text{vi) } y &= 0 \\ 5x-9 &= 0 \\ x &= \frac{9}{5} \end{aligned}$$

$$\begin{aligned} \text{vii) } (0, 2) \\ \text{dist} &= \frac{|5 \times 0 + -4 \times 2 - 9|}{\sqrt{5^2 + 4^2}} \\ &= \frac{17}{\sqrt{41}} \end{aligned}$$

$$\text{viii) } D(-3, -6)$$

$$\begin{aligned} \text{ix) } &\sqrt{41} \times \frac{17}{\sqrt{41}} \\ &= 17 \text{ sq units} \end{aligned}$$

$$\begin{aligned} \text{x) } E(x, y) \\ \frac{x+1}{2} &= 0 \quad \frac{y-1}{2} = 2 \\ E &(-1, 5) \end{aligned}$$

$$\begin{aligned} \text{xi) } (0, 2) \quad m &= \tan 135^\circ \\ &= -1 \end{aligned}$$

$$\begin{aligned} y-2 &= -1(x-0) \\ y &= 2-x \\ x+y-2 &= 0. \end{aligned}$$

$$\begin{aligned} \text{Q2) a) } y-3 &= 3(x-2) \\ y-3 &= 3x-6 \\ y &= 3x-3. \end{aligned}$$

$$\text{b) } m = \frac{5}{3}$$

$$\begin{aligned} \therefore \perp \text{ gradient} &= -\frac{3}{5} \\ y+1 &= -\frac{3}{5}(x-2) \\ 5y+5 &= -3x+6 \\ 3x+5y-1 &= 0 \end{aligned}$$

$$\begin{aligned} \text{c) } 3x-4y-1+k(2x+3y-5) &= 0 \\ \text{passes through } (1, 2) \\ 3-8-1+3k &= 0 \\ 3k &= 6 \end{aligned}$$

$$k = 2.$$

$$\begin{aligned} 3x-4y-1+2(2x+3y-5) &= 0 \\ 3x-4y-1+4x+6y-10 &= 0 \\ 7x+2y-11 &= 0 \end{aligned}$$

Q3)
a) tangent if distance from centre of the circle equals the radius (4)

$$(0, 0) \quad 3x+4y-20=0$$

$$\begin{aligned} d &= \frac{|0+0-20|}{\sqrt{3^2+4^2}} \\ &= \frac{20}{5} \\ &= 4 \end{aligned}$$

= radius of the circle. $\therefore$ a tangent

$$\begin{aligned} 3x-4y+9 &= 0, \\ 3x-4y-20 &= 0 \\ (0, 5) &\text{ lies on the } 2^{\text{nd}} \text{ line.} \end{aligned}$$

$$\begin{aligned} \therefore d &= \frac{|3 \times 0 + -4 \times 5 + 9|}{\sqrt{3^2 + 4^2}} \\ &= \frac{29}{5} \text{ units.} \end{aligned}$$

$$Q4) 3\sqrt{5} - \frac{1}{\sqrt{5}-2} = a+b\sqrt{5}$$

LHS

$$= 3\sqrt{5} - \frac{1}{\sqrt{5}-2} \times \frac{\sqrt{5}+2}{\sqrt{5}+2}$$

$$= 3\sqrt{5} - \frac{\sqrt{5}+2}{5-4}$$

$$= 3\sqrt{5} - \sqrt{5} - 2$$

$$= 2\sqrt{5} - 2$$

$$\therefore a = -2, b = 2$$

$$b) x^2 \leq 5x$$

$$x^2 - 5x \leq 0$$

$$x(x-5) \leq 0$$

$$\begin{array}{c} \text{---} 0 \quad 5 \text{---} \end{array}$$

$$\text{test } x=1, 1-5 \leq 0 \text{ T.}$$

$$\therefore 0 \leq x \leq 5$$

$$c) |x-1| \geq 4$$

$$4 \leq x-1 \quad \text{or} \quad x-1 \leq -4$$

$$5 \leq x \quad \text{or} \quad x \leq -3$$

$$d) a^2 - b^2 + 2bc - c^2$$

$$= a^2 - (b^2 - 2bc + c^2)$$

$$= a^2 - (b-c)^2$$

$$= (a+(b-c))(a-(b-c))$$

$$= (a+b-c)(a-b+c)$$

$$e) \frac{4^x}{16} = 8^{x+y}$$

$$4^x = 16 \cdot 8^{x+y}$$

$$2^{2x} = 2^4 \cdot 2^{3x+3y}$$

$$2^{2x} = 2^{3x+3y+4}$$

$$\therefore 2x = 3x+3y+4$$

$$x+3y = -4 \quad \dots (1)$$

$$2^{2x+y} = 128$$

$$2^{2x+y} = 2^7$$

$$\therefore 2x+y = 7 \quad \dots (2)$$

$$(1) \times 2, 2x+6y = -8 \quad \dots (3)$$

$$(3)-(2) \quad 5y = -15$$

$$y = -3$$

$$\text{Sub } y = -3 \text{ into (1)}$$

$$x - 9 = -4$$

$$x = 5$$